

Computational Physics: Materials Science

Exercise 6

Diffusion coefficient in liquid molecular dynamics simulations

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Authors:	Stefanus Michael 5976298 Microsystems Engineering



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This exercise extends the Ex. 5 molecular-dynamics code to the study of particle diffusion. The Morse potential of the Na₂₇ cluster is replaced by the Lennard-Jones (12-6) pair potential, and the system is changed from a small bound cluster to a periodic Lennard-Jones fluid of $N = 125$ argon-like particles. The goal is to compute the self-diffusion coefficient D by two independent routes — the Einstein relation from the mean-squared displacement (MSD) and the Green-Kubo relation from the velocity autocorrelation function (VACF) — and to study how D scales with density, system size and temperature. The Atomic Simulation Environment (ASE) drives the dynamics; the numba force kernel is reused from Ex. 4/5 with the Morse force replaced by the Lennard-Jones force. An AI assistant (Claude) helped with the ASE/NVE integration, the FFT-based estimators and plot generation; the physical interpretation is my own.

Implementation

Potential

The Lennard-Jones (12-6) potential models the dispersion ($-r^{-6}$) attraction and the Pauli ($+r^{-12}$) repulsion of closed-shell atoms,

$$V_{\text{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right], \quad \mathbf{F}_i = - \sum_{j \neq i} \frac{dV_{\text{LJ}}}{dr} \frac{\mathbf{r}_{ij}}{r_{ij}}. \quad (1)$$

It is the standard model for rare-gas and simple fluids. This differs from the Ex. 5 Morse potential, whose exponential form is tailored to cohesive metallic/covalent bonding (the Na cluster); the Lennard-Jones r^{-6} tail instead reproduces the physical van-der-Waals attraction of a noble-gas fluid. The potential is evaluated with the minimum-image convention and shifted to zero at the cutoff $r_{\text{cut}} = 2.5\sigma$, so that $V_{\text{LJ}}(r_{\text{cut}}) = 0$ and no energy jump occurs at the cutoff.

Diffusion estimators

Einstein relation. In the long-time diffusive regime the MSD grows linearly with time, and in three dimensions

$$\text{MSD}(\tau) = \langle |\mathbf{r}_i(t_0 + \tau) - \mathbf{r}_i(t_0)|^2 \rangle_{i,t_0}, \quad D = \lim_{\tau \rightarrow \infty} \frac{\text{MSD}(\tau)}{6\tau}. \quad (2)$$

In practice D is obtained from a straight-line fit $\text{MSD} = 6D\tau + c$ in the diffusive window (here the lag interval $[0.2, 0.5] \tau_{\text{max}}$, long enough to be diffusive yet short enough to be well averaged).

Green-Kubo relation. Equivalently, D is the time integral of the velocity autocorrelation function,

$$D = \frac{1}{3} \int_0^\infty \langle \mathbf{v}_i(0) \cdot \mathbf{v}_i(t) \rangle_i dt. \quad (3)$$

Two refinements required by the sheet. (i) *Statistical stability.* Rather than a single time origin $t_0 = 0$, Eq. (2) is averaged over *all* available time origins t_0 (and all particles). This sliding-window average uses every frame as a reference and sharply



reduces the variance of the estimate, especially at long lag. (ii) *Speed*. The multiple-origin sum is a correlation; evaluated directly it costs $O(T^2)$ in the number of frames T . Using the fast Fourier transform (Wiener–Khinchin theorem) it is reduced to $O(T \log T)$. The same FFT machinery is reused for the VACF in Eq. (3). 

Code stack

The unit system, shifted cutoff and numba force kernel are inherited from Ex. 5; the Morse force is replaced by Eq. (1). New to Ex. 6:

- **Lennard-Jones ASE Calculator.** A subclass of `ase.calculators.Calculator` calls the numba Lennard-Jones kernel and writes energy and forces into the `results` dictionary, so the ASE integrators use the exact shifted-cutoff potential of Eq. (1).
- **Two-stage dynamics.** Equilibration uses a *strong* Langevin thermostat ($\gamma = 0.01 \text{ fs}^{-1}$, relaxation time $\approx 100 \text{ fs}$) so the simple-cubic seed melts and the temperature locks onto T_0 within a fraction of a picosecond. Production is then run in the *NVE* ensemble (velocity Verlet).
- **Why NVE production.** A thermostat that damps momentum (such as Langevin) is not momentum-conserving and therefore (a) biases the measured transport coefficient and (b) screens the long-range hydrodynamic correlations responsible for the finite-size effect studied in Sec. (e). Running production in NVE removes both problems and additionally eliminates the spurious centre-of-mass drift; the net momentum is zeroed once before production starts.
- **MSD/VACF analysis.** FFT-based multiple-origin MSD and VACF, the Einstein straight-line fit, the cumulative Green–Kubo integral, and parameter sweeps over density, box size and temperature (each sweep point averaged over independent random seeds for error bars).

Simulation parameters

N	$\rho^* = \rho\sigma^3$	$L [\text{\AA}]$	$\Delta t [\text{fs}]$	N_{equil}	N_{prod}	$\sigma [\text{\AA}]$	$r_{\text{cut}} [\text{\AA}]$
125	0.5	21.48	2	10000	10000	3.41	8.53

Table 1: Simulation parameters, following Table 1 of the sheet: $N = 125$, $\Delta t = 2 \text{ fs}$, $N_{\text{equil}} = N_{\text{prod}} = 10000$, $m = 39.95 \text{ u}$ (argon), $T_0 = 300 \text{ K}$, $\sigma = 0.341 \text{ nm} = 3.41 \text{\AA}$, $\rho^* = 0.5$. The well depth is read as $\varepsilon = 0.3 k_B T_0 = 7.76 \text{ meV}$ (the one entry of Table 1 open to interpretation). Equilibration uses a strong Langevin thermostat ($\gamma = 0.01 \text{ fs}^{-1}$); production is NVE.

The equilibration is monitored to confirm that $N_{\text{equil}} = 10000$ steps (20 ps) is sufficient: Fig. 1 shows that, after a melting transient of $\lesssim 0.5 \text{ ps}$, both the temperature and the total energy fluctuate about stationary means with no residual drift. The production run holds $\langle T \rangle \approx 307 \text{ K}$ with $\sigma_T \approx 9 \text{ K}$.

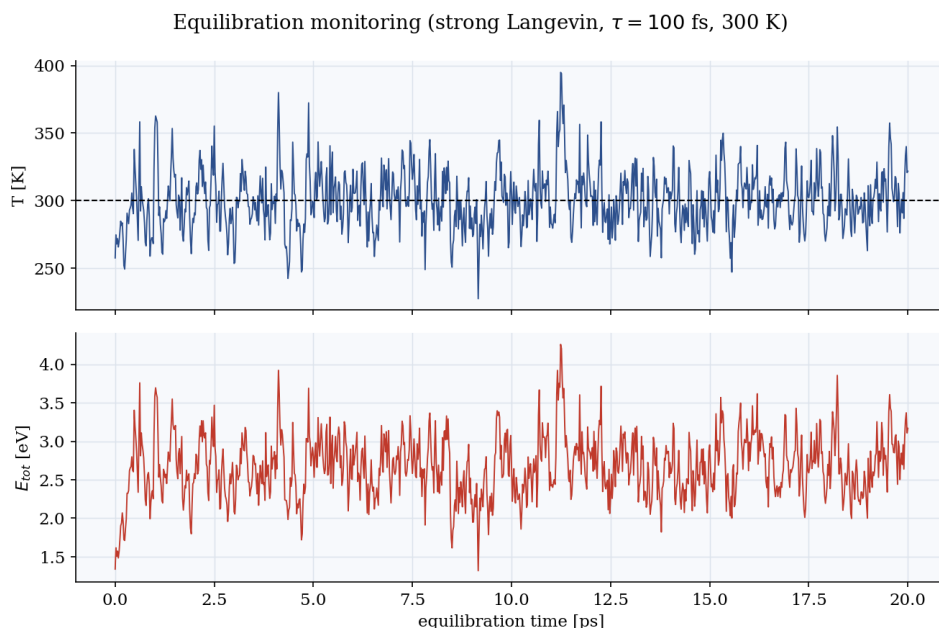


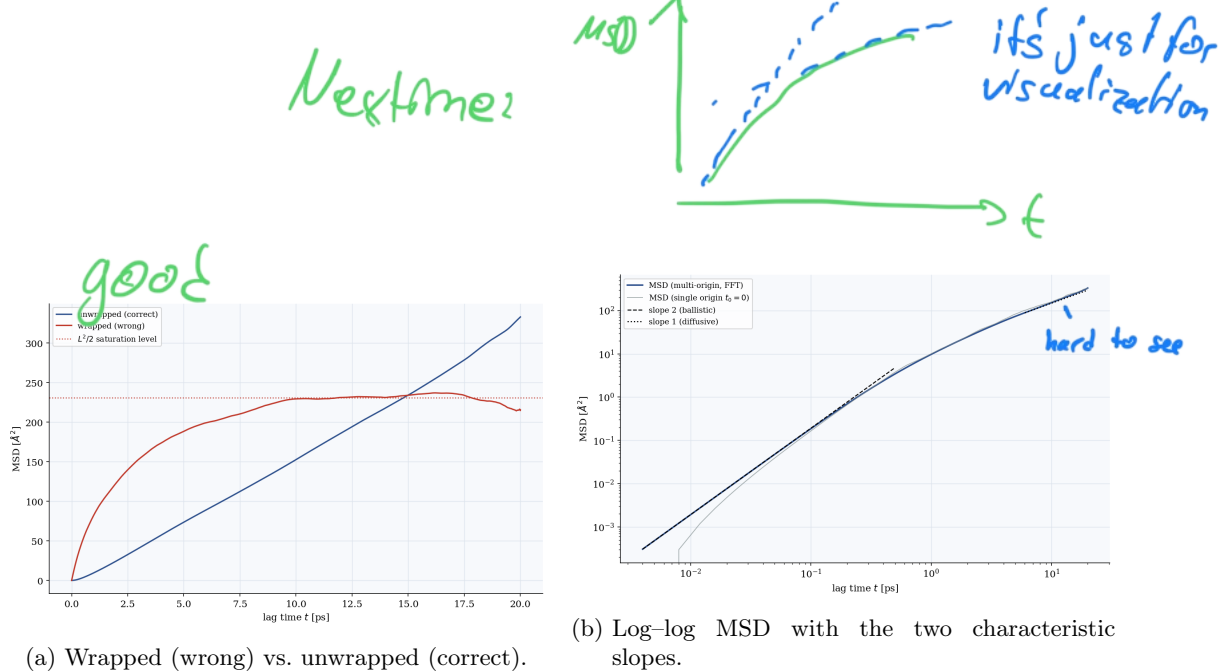
Figure 1: Equilibration monitoring at 300 K (strong Langevin, $\tau = 100$ fs). Top: instantaneous temperature about the target (dashed). Bottom: total energy. Both reach a stationary plateau within $\lesssim 0.5$ ps, confirming the equilibration window is more than sufficient before production begins.

Analysis

(a) Mean-squared displacement: wrapped vs. unwrapped, and the two regimes

The wrapped-coordinate mistake. Under periodic boundaries, wrapping folds a particle's coordinate back into the box whenever it crosses a face. This introduces artificial jumps of size L into the trajectory, so a displacement computed from wrapped coordinates can never exceed the box scale. The wrapped MSD therefore *saturates* instead of growing: for two uncorrelated points uniformly distributed in a cubic box the mean-squared separation is $L^2/2$, and Fig. 2a shows the wrapped curve levelling off exactly at that value ($L^2/2 \approx 231 \text{ \AA}^2$). No diffusion coefficient can be extracted from it. The correct MSD uses the *unwrapped* positions (ASE stores these directly), which grow linearly in the diffusive regime.





(a) Wrapped (wrong) vs. unwrapped (correct).
 Figure 2: (a) The wrapped MSD saturates at the geometric ceiling $L^2/2$, while the unwrapped MSD grows linearly. (b) On log-log axes the unwrapped MSD shows the ballistic regime (slope 2) at short times and the diffusive regime (slope 1) at long times; the multiple-origin (FFT) estimate is far smoother than the single-origin one, especially at long lag.

The two regimes. On log-log axes (Fig. 2b) the unwrapped MSD exhibits the two expected power laws: at short times $\text{MSD} \propto t^2$ (slope 2, the *ballistic* regime of free flight before the first collision), crossing over near ~ 0.5 ps to $\text{MSD} \propto t$ (slope 1, the *diffusive* regime). The clean slope-1 region at long times is where D is extracted and serves as the validation that the simulation behaves as expected. The multiple-origin (FFT) curve is visibly smoother than the single-origin estimate, illustrating the statistical-stability gain of refinement (i).

(b) Einstein diffusion coefficient and dependence on run length

A straight-line fit in the diffusive window (Fig. 3) gives the reference value

$$D_{\text{Einstein}} = 2.61 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \quad (\rho^* = 0.5, T = 300 \text{ K}, N = 125). \quad (4)$$

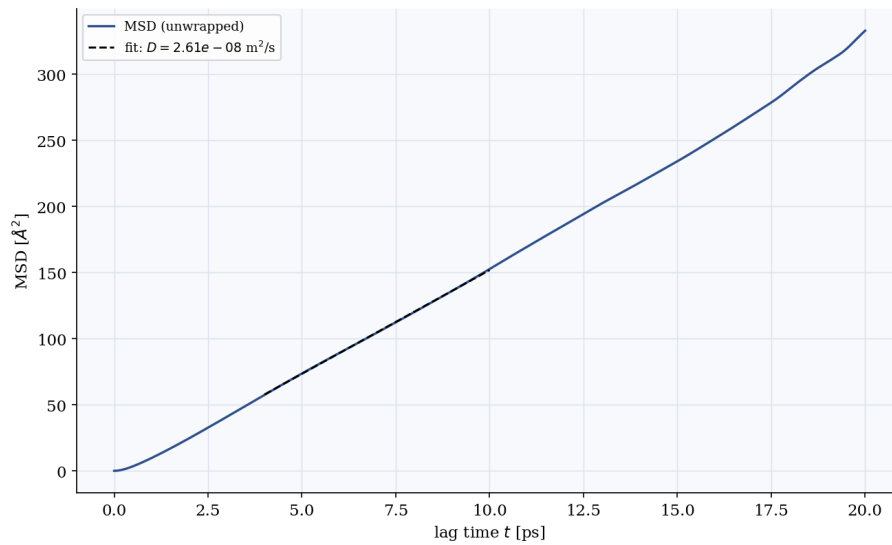


Figure 3: Unwrapped MSD of the main run with the linear Einstein fit (dashed) in the diffusive window. The slope yields $D = 6^{-1} \text{ d MSD}/\text{d}t$.



Repeating the measurement for different equilibration/production lengths (Table 2) probes the convergence of D . The shortest run clearly underestimates D , because its production window is too short to develop a long, well-averaged diffusive regime. Beyond that, the single-seed values scatter by $\sim 15\%$ around $D \approx 2.6 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$ without a clean monotonic trend: a single trajectory carries a large statistical uncertainty. The longest run ($N_{\text{prod}} = 30000$) returns $2.59 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$, consistent with the main run and with the seed-averaged values used in Secs. (d)–(f). The conclusion is twofold: production must be long enough to reach a well-sampled diffusive window, and seed averaging is required for a converged estimate — which is why the parameter sweeps below average over several independent seeds.

N_{equil}	N_{prod}	$D [10^{-8} \text{ m}^2 \text{ s}^{-1}]$	note
1000	2000	2.15	shortest: underestimates D
5000	5000	2.70	
10000	10000	3.05	Table 1 setting (single seed)
10000	30000	2.59	longest: closest to seed-averaged value

Table 2: Einstein D for different equilibration/production lengths (single seed each). The shortest run underestimates D ; the remaining values scatter within the single-seed statistical spread ($\sim 15\%$), with the longest run closest to the seed-averaged result.

(c) Green–Kubo diffusion coefficient and its convergence

The velocity autocorrelation function (Fig. 4, top) decays to zero within $\sim 2 \text{ ps}$ with only a weak negative tail, as expected for a moderately dense fluid. Its running integral, Eq. (3), plateaus (Fig. 4, bottom) at $D_{\text{GK}} \approx 2.81 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$, in good agreement (within $\sim 7\%$) with the Einstein value $2.61 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$ (dashed). The agreement of the two independent estimators is the principal validation of the method.



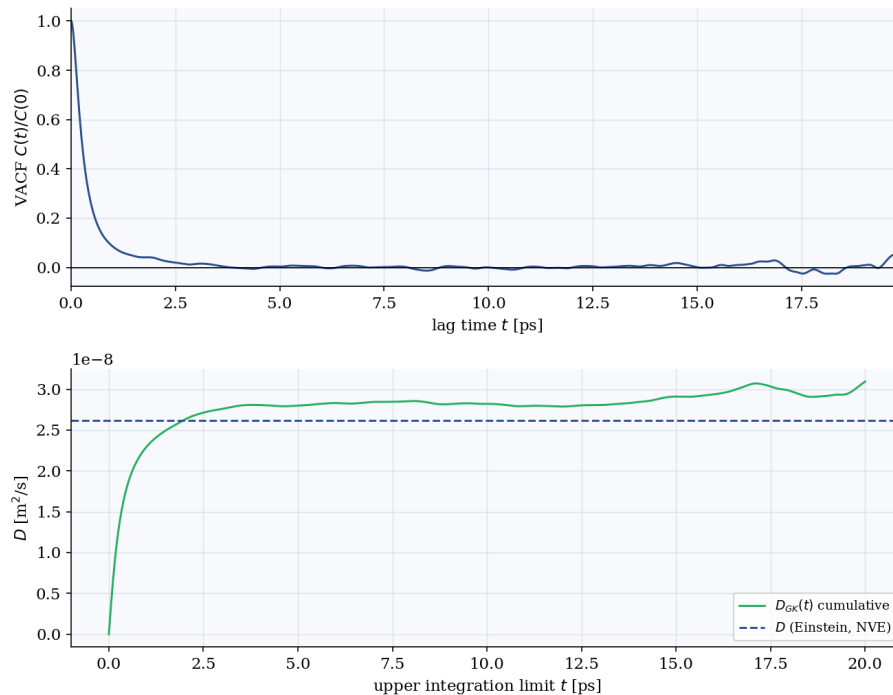


Figure 4: Top: normalised VACF $C(t)/C(0)$. Bottom: cumulative Green–Kubo integral $D_{\text{GK}}(t)$ vs. upper integration limit, with the Einstein value (dashed) for comparison. The integral plateaus once the VACF has decayed and then drifts at large t as statistical noise accumulates.

Why Green–Kubo is difficult to converge. The integral formally runs to $t \rightarrow \infty$, but at large lag the VACF is dominated by statistical noise (progressively fewer independent time origins contribute) and decays only slowly through its hydrodynamic tail. The cumulative integral therefore plateaus first and then drifts (visible beyond ~ 15 ps in Fig. 4); D_{GK} must be read from the plateau before the noise takes over. The Einstein estimator, being a fit to an integrated quantity, is more robust here. ✓

Thermostat implications. Because a momentum-damping thermostat biases transport, production is run in NVE. A control run with weak Langevin production at the same state point gives $D_{\text{Langevin}}/D_{\text{NVE}} = 0.81$, i.e. the thermostat lowers D by $\sim 20\%$; more importantly, only the momentum-conserving NVE dynamics retain the hydrodynamic finite-size effect verified in Sec. (e).

(d) Density dependence

At fixed temperature the diffusion coefficient decreases monotonically with density (Fig. 5): from $D \approx 5.43 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$ at $\rho^* = 0.3$ to $D \approx 1.10 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$ at $\rho^* = 0.8$. Physically, a higher density shortens the mean free path and increases the collision frequency and cage effect, so particles diffuse more slowly.

Especially early time values are modified by stochastic kicks



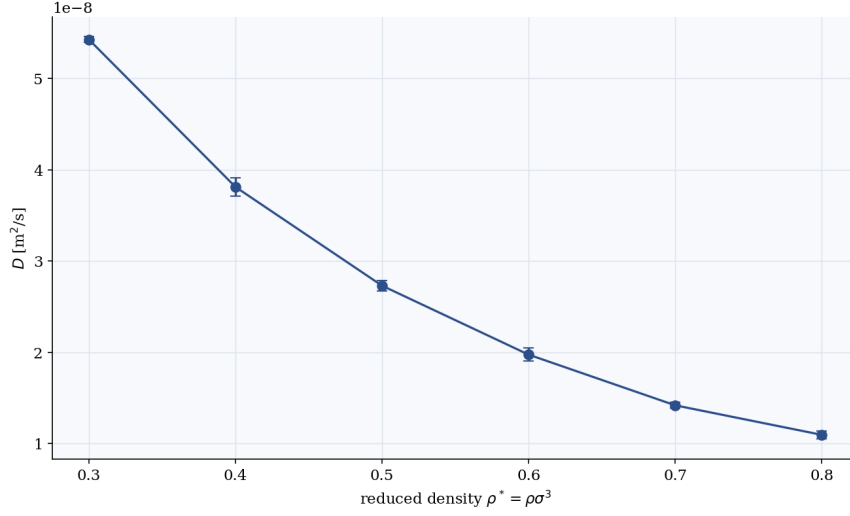


Figure 5: Self-diffusion coefficient vs. reduced density $\rho^* = \rho\sigma^3$ at $T = 300 \text{ K}$ (error bars over independent seeds). D decreases monotonically with density.

(e) System-size dependence and finite-size extrapolation

Keeping ρ^* fixed and varying the box from $N = 64$ to $N = 512$ particles, the measured D increases with box size L (equivalently, decreases with $1/L$, Fig. 6). This is the hydrodynamic finite-size effect: under periodic boundaries the tagged particle interacts with its own periodic images through the solvent backflow, which slows it down in a smaller box. The Yeh–Hummer correction predicts a linear dependence on $1/L$,

$$D_{\text{PBC}}(L) = D_{\infty} - \frac{\xi k_B T}{6\pi\eta L}, \quad \xi = 2.837297, \quad (5)$$

with η the shear viscosity. A weighted linear fit of D vs. $1/L$ extrapolates to the infinite-system value

$$D_{\infty} = 2.89 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \quad (1/L \rightarrow 0). \quad (6)$$

The negative slope is consistent with Eq. (5); identifying it with $\xi k_B T / (6\pi\eta)$ yields an effective shear viscosity $\eta \approx 0.13 \text{ mPa s}$, of the same order as liquid argon ($\sim 0.2 \text{ mPa s}$), an order-of-magnitude sanity check rather than a precision measurement. The base system of Table 1 is $N = 125$; the smaller and larger boxes bracket it to give a wider $1/L$ lever arm for the extrapolation.



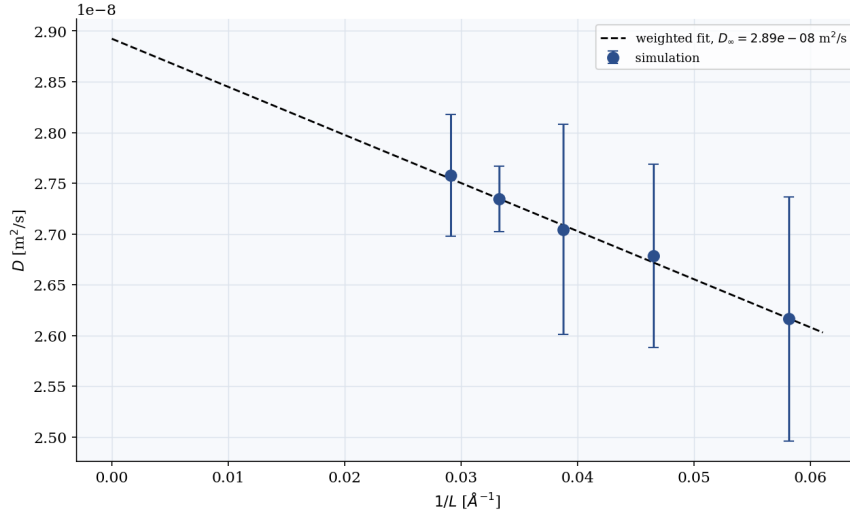


Figure 6: Self-diffusion coefficient vs. inverse box length $1/L$ at fixed $\rho^* = 0.5$, $T = 300 \text{ K}$ (NVE production, error bars over independent seeds). The linear Yeh–Hummer trend extrapolates to the infinite-system value D_∞ at $1/L \rightarrow 0$.

(f) Temperature dependence: Arrhenius behaviour

The diffusion coefficient increases with temperature. Plotting $\ln D$ against $1/\langle T \rangle$ (Fig. 7) gives a straight line, i.e. D follows the Arrhenius law

$$D(T) = D_0 \exp\left(-\frac{E_A}{k_B T}\right), \quad (7)$$

with an activation energy $E_A = 17.7 \text{ meV}$ and prefactor $D_0 = 5.44 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$ (fit performed against the *measured* mean temperatures $\langle T \rangle$, not the nominal targets). The fit is approximately Arrhenius over this range; a slight curvature appears at the cold end, where the fluid is more liquid-like, whereas at the hottest points it is closer to a supercritical gas.

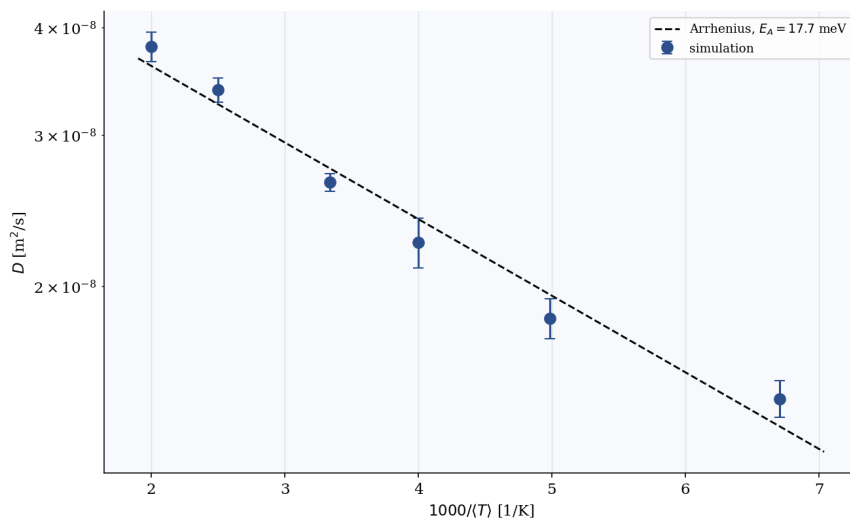


Figure 7: Arrhenius plot of D vs. $1000/\langle T \rangle$ on a logarithmic D axis (error bars over independent seeds). The straight-line fit (dashed) gives the activation energy E_A .



If you use unwrapped it is hard to see fluid. Do wrapped

Ovito visualisation

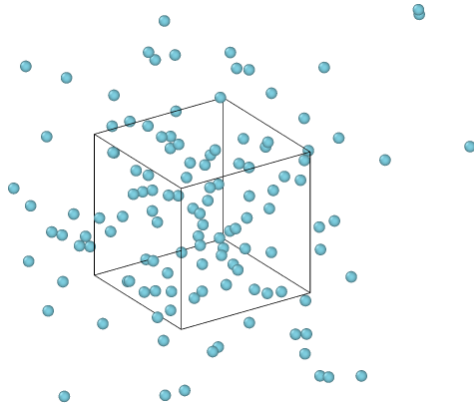


Figure 8: Snapshot from Ovito of the $N = 125$ system at $\rho^* = 0.5$, $T = 300$ K, taken from the production trajectory. The particles fill the periodic box as a disordered, mobile fluid (no bound cluster), confirming that the simulation produces a proper liquid for which the self-diffusion coefficient is well defined.

Summary

- **Two estimators agree.** The Einstein MSD fit gives $D = 2.61 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$ at $\rho^* = 0.5$, $T = 300$ K; the Green–Kubo VACF integral plateaus at $\approx 2.81 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$, validating the method.
- **MSD methodology.** Wrapped coordinates give a wrong, saturating MSD ($L^2/2$ ceiling); the correct unwrapped MSD shows the ballistic (slope 2) and diffusive (slope 1) regimes. Multiple time origins stabilise the estimate and the FFT reduces its cost from $O(T^2)$ to $O(T \log T)$.
- **Convergence.** The shortest production run underestimates Einstein D ; longer runs scatter within the single-seed spread, with the longest closest to the seed-averaged value (Table 2). Green–Kubo is harder to converge because of the noisy long-time tail of the VACF integral.
- **Scaling laws.** D decreases monotonically with density; increases with box size following the Yeh–Hummer $1/L$ law, extrapolating to $D_\infty = 2.89 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$; and follows the Arrhenius law in temperature with $E_A = 17.7 \text{ meV}$.
- **Thermostat choice.** Production is run in NVE so that the thermostat neither biases D (weak Langevin gives only 0.81 of the NVE value) nor screens the hydrodynamic finite-size effect; equilibration uses a strong Langevin thermostat to reach the target state quickly.
- **Limitations.** The finite-size extrapolation and the effective viscosity ($\approx 0.13 \text{ mPa s}$) are order-of-magnitude robust but limited by statistics; the Arrhenius fit is only approximate as the fluid changes character with temperature over the sampled range.

Very nice! No notes from my side!

